Tandem Bicycle

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The problem

- Pair red and blue riders; a tandem's speed is the max of its riders.
- Return the max (fastest=true) or min possible total speed.

Intuition

- Speed is the max of the two riders.
- For maximum total, pair the fastest with the slowest; for minimum, pair like with like.

Approach

- 1. Sort both lists
- 2. For fastest total, pair one ascending with the other descending
- 3. For slowest, pair in the same order
- 4. Sum the max of each pair

Java solution

```java
public class TandemBicycle {
    public static int tandemBicycle(int[] red, int[] blue, boolean fastest) {
        Arrays.sort(red);
        Arrays.sort(blue);
        int n = red.length, total = 0;
        for (int i = 0; i < n; i++) {
            int b = fastest ? blue[n - 1 - i] : blue[i];
            total += Math.max(red[i], b);
        }
        return total;
    }
}
```

Complexity

- Time: O(n log n)
- Space: O(1)

Recap

- Speed is the max of the two riders.
- Complexity: O(n log n)